

3 The Nature of Molecules

3.1 Matter

Matter is defined as anything that has mass¹ (an amount of matter in an object) and occupies space² (which is measured as volume³).

- Particles, from smallest to largest
 1. Subatomic particles
 - Electrons⁴
 - Protons⁵
 - Neutrons⁶
 2. Atoms
 3. Molecules
 4. Macromolecules
- Origin of matter
 1. Big Bang⁷, about 13.7 billion years ago
 2. Hydrogen⁸, helium⁹
 3. Heavier elements formed in suns, super nova
 - Earth's matter predates formation of sun, 4.5 billion years ago
- All matter consists of atoms, which are composed of : electrons, protons, neutrons

3.2 The atom

- Example: Hydrogen
 - The simplest element
 - One proton (+)
 - One electron in orbit (-)
- Built by adding one proton (and one electron) at a time
- Number of protons determines atomic number and number of electrons
- Neutrons
 - Neutral charge

1 <http://en.wikipedia.org/wiki/mass>
2 <http://en.wikipedia.org/wiki/space%23Physics>
3 <http://en.wikipedia.org/wiki/volume>
4 <http://en.wikipedia.org/wiki/Electrons>
5 <http://en.wikipedia.org/wiki/Protons>
6 <http://en.wikipedia.org/wiki/Neutrons>
7 <http://en.wikipedia.org/wiki/Big%20Bang>
8 <http://en.wikipedia.org/wiki/Hydrogen>
9 <http://en.wikipedia.org/wiki/Helium>

- Contribute mass
- May decay
- Oxygen¹⁰
 - 8 protons (mass)
 - 8 electrons
 - 8 neutrons (mass)

3.3 Mass and isotopes

- Atomic mass
 - Sum of masses of protons and neutrons
 - Measured in daltons or AMU (Atomic Mass Unit)
 - An AMU is 1/12 the mass of Carbon-12
 - proton ~1 AMU or dalton
 - 6.024×10^{23} daltons/gram
- Atoms with same atomic number belong to same element
- Isotopes
 - Same atomic number but different atomic mass
 - Some are radioactive
 - Uses of isotopes
 - Radioactive: ^3H , ^{14}C , ^{32}P , ^{35}S
 - Tracers in biochemical reactions
 - Detection of molecules in recombinant DNA technology (genetic engineering)
 - Half-life: dating of rocks, fossils
 - Non-radioactive (N, C, O)
 - Diet of organisms (including fossils)
 - Biochemical tracers

3.4 Electrons

- Negative charge
- Held in orbit about nucleus by attraction to positively charged nucleus
- Atom may gain or lose electron, altering charge
 - Cation: loses electron, positive charge
 - Na^+
 - Anion: gains electron, negative charge
 - Cl^-
- Determine chemical properties of atoms
 - Number
 - Energy level

¹⁰ <http://en.wikipedia.org/wiki/Oxygen>

3.5 Chemical bonds

- Form molecules
- Enzymes: make, break, rearrange chemical bonds in living systems
- Ionic
- Covalent
 - Sharing of one or more pairs of electrons
 - Called single, double, or triple
 - No net charge (as in ionic bonds)
 - No free electrons
 - Give rise to discrete molecules
 - Hydrogen

3.6 Chemical reactions

- Formation and breaking of chemical bonds
- Shifting arrangement of atoms
- Reactants -> products
- Reactions are influenced by:
 - Temperature
 - Concentration of reactants, products
 - Presence of catalysts (enzymes)
- Oxidation:reduction

3.7 Water

- Essential for life
- ~75% earth's surface is water
- Life evolved in water
- Solvent for many types of solutes
- High specific heat
- High polarity
 - Creates a slightly negative Oxygen and a Slightly positive hydrogen
 - allows formation of Hydrogen Bonds

3.7.1 Hydrogen bonding

- A type of polar interaction
- Critical for:
 - Protein structure
 - Enzymatic reactions
 - Movement of water in plant stems
- Weak and transient
- Powerful cumulative effect

- Solubility of many compounds
- Cohesion (capillary action)
- Lower density of ice
- Formed between molecules other than water
 - Protein structure
 - DNA¹¹, RNA¹² structure

Water organizes nonpolar molecules

- Nonpolar molecules: no polarity (+/-) charges
- Hydrophobic: exclude water because they don't form hydrogen bonds with it
- Consequences:
 - Membranes
 - Protein structure
- Hydrophilic: polar substances associate with water

Ionization of water: $\text{H}_2\text{O} \rightarrow \text{H}^+ + \text{OH}^-$

- Forms a Hydrogen ion (H^+), hydroxide ion (OH^-)
- Due to spontaneous breakage of covalent bond
- At 25°C, 1 liter of water contains 10^{-7} moles of H^+ ions: 10^{-7} moles/liter

pH

- A convenient way of indicating H^+ concentration
- $\text{pH}^{13} = -\log[\text{H}^+]$
- For water, $\text{pH} = -\log[10^{-7}] = 7$
- Since for each H^+ in pure water, there is one OH^- , pH of 7 indicates neutrality
- Logarithmic scale

Buffer

- Reservoir for H^+
- Maintains relatively constant pH over buffering range

This text is based on notes very generously donated by Dr. Paul Doerder, Ph.D., of the Cleveland State University.

¹¹ <http://en.wikipedia.org/wiki/DNA>

¹² <http://en.wikipedia.org/wiki/RNA>

¹³ <http://en.wikipedia.org/wiki/pH>